Does Expected Goals (xG) Predict Actual Goals?

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Introduction

Soccer is the most popular sport in the world and continues to grow through the use of data and analytics. Professional clubs, coaches, and analysts now use advanced statistics to better understand team performance, player development, and match outcomes. One of the most important statistics in modern soccer analytics is expected goals, commonly known as xG.

Expected goals is a metric that estimates the probability of a shot becoming a goal based on factors such as shot location, angle, and scoring situation. Instead of only looking at the final score, xG helps measure the quality of scoring opportunities created during a match. This gives analysts a deeper understanding of performance and helps reduce some of the randomness that naturally exists in soccer.

Although xG is widely used in professional soccer, it is still important to evaluate how well it predicts actual goals scored. The purpose of this project is to examine the relationship between expected goals and actual goals using match level data from the Understat database. This study uses 24,580 observations from professional soccer matches and applies multiple statistical models, including linear regression, polynomial regression, Poisson regression, negative binomial regression, ridge regression, and LASSO regression, to determine how accurately xG predicts goal scoring performance.

The main research question for this project is: Does expected goals predict actual goals scored in professional soccer? This study aims to provide statistical evidence that helps answer this question while showing how advanced analytics can improve the understanding of soccer performance.

Literature Review

Expected Goals Models in Sports Analytics

Expected goals (xG) has become an important metric in modern soccer analytics because it estimates the probability that a shot will result in a goal based on factors such as shot distance, angle, and match context. In a peer reviewed study, **James Mead, Anthony O'Hare, and Paul McMenemy (2023)** examined ways to improve xG models in professional soccer by applying machine learning techniques and additional contextual variables. The researchers found that xG was a stronger predictor of future team performance than traditional statistics such as goals scored or total shots. The authors concluded that xG is a valuable analytical tool for evaluating offensive efficiency and predicting success in soccer. This research supports the use of xG as a key predictor in this senior project.

Machine Learning in Soccer Sports Analytics

The authors **James Hewitt** and **Oktay Karakuş** (2023) examined how machine learning can improve soccer performance analysis through expected goals modeling. Using event data from **StatsBomb**, the researchers developed player and position adjusted xG models to better evaluate scoring opportunities in soccer. Their results showed that the proposed model achieved strong predictive accuracy and closely matched industry benchmark xG values. The study concluded that machine learning can improve player evaluation, recruitment, and tactical decision making in soccer. This research supports the use of advanced statistical models in this senior project.

Tracking and Event Data in Soccer Performance Analysis

The authors **Steffen Lang, Raphael Wild, Alexander Isenko, and Daniel Link (2022)** examined how tracking data and machine learning can improve soccer performance analysis. The researchers analyzed spatiotemporal player tracking data from **102 professional matches in the German Bundesliga** and tested multiple machine learning models to identify in-game match events. Their results showed that the **AdaBoost model achieved 92% prediction accuracy** and an **F1 score of 0.93**, demonstrating that tracking data can accurately detect important match situations. The authors concluded that combining tracking and event data can improve tactical analysis, event detection, and performance evaluation in soccer. This research supports the use of advanced data in soccer analytics.

Data and Methodology

This study used match level soccer data obtained from **Understat**, a widely used database that provides advanced statistical information for professional soccer matches across major European leagues. The dataset included matches from top European competitions and contained **24,580 observations**, where each row represented one team in one match. This large sample size provided a strong foundation for statistical analysis and allowed for more reliable model estimation. The purpose of using match level data was to examine whether expected goals can

accurately predict actual goals scored while accounting for additional offensive and defensive performance factors.

The dependent variable in this study was **goals scored (scored)**, which measured the total number of goals scored by a team in
a single match. Because goals are the most direct measure of offensive success in soccer, this variable was selected as the response variable for all statistical models.

Several predictor variables were included in the analysis. The main predictor was **expected goals (xG)**, which estimates the probability of scoring based on shot quality. Additional offensive and defensive variables included **expected goals difference (xG_diff)**, **expected goals against difference (xGA_diff)**, and **non penalty expected goals difference (npxGD)**. These variables were included to capture overall team performance and scoring efficiency beyond basic goal totals.

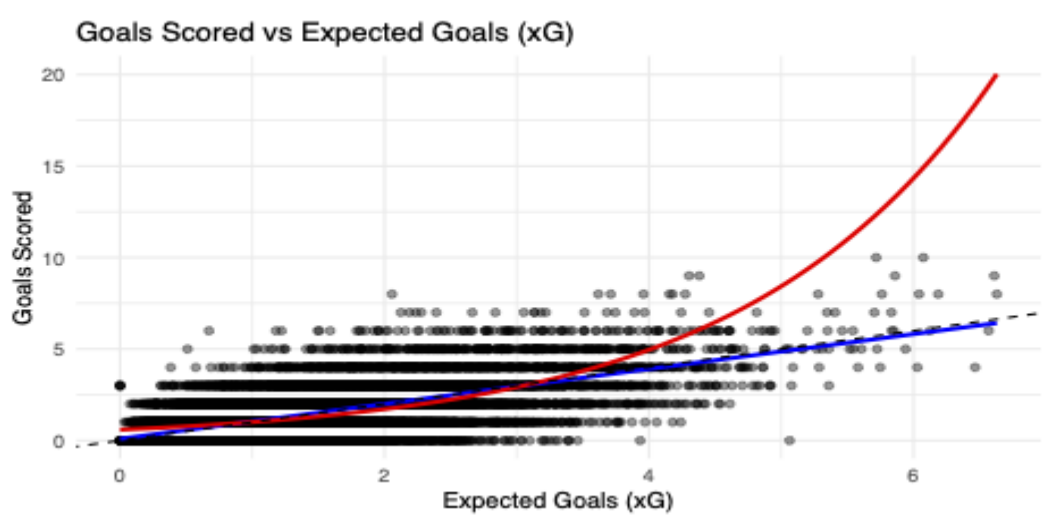
This study also included pressing related variables: **ppda_coef**, **oppda_coef**, **ppda_att**, **ppda_def**, **oppda_att**, and **oppda_def**. These variables measure defensive intensity and opponent pressure by tracking how frequently teams allow passes before applying defensive actions. Pressing metrics provide valuable information about team tactics, defensive organization, and how pressure may influence goal scoring opportunities.

All statistical analyses were performed using **R**. Several packages were used throughout the analysis, including **tidyverse** for data cleaning and visualization, **MASS** for count regression models, and **glmnet** for ridge and LASSO regression models. These tools allowed multiple statistical approaches to be applied and compared throughout the study.

Exploratory Data Analysis

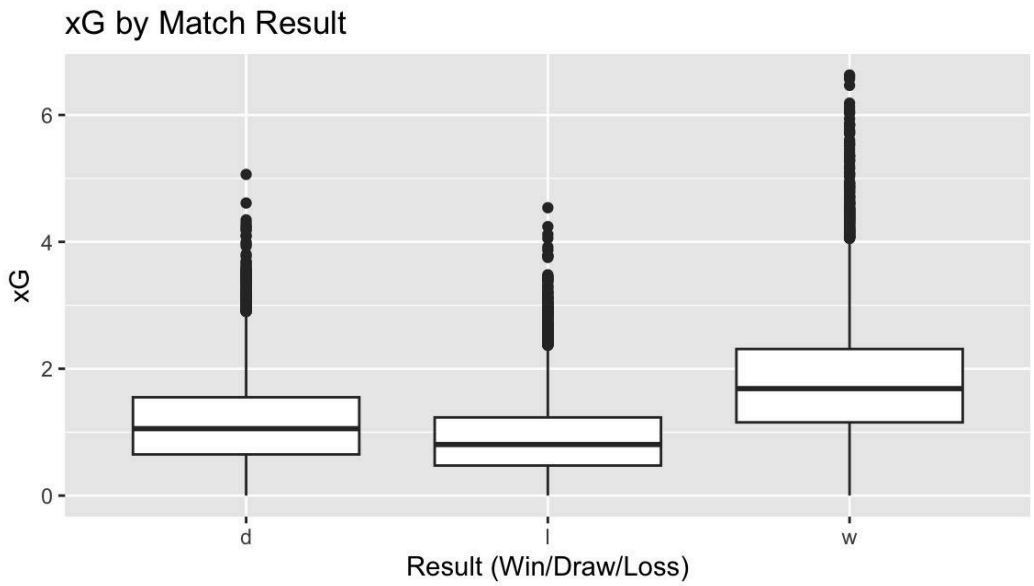
When I fit the statistical models, exploratory data analysis was performed to better understand the relationships between expected goals, goals scored, and other match characteristics in the dataset. Several visualizations were created to identify patterns, detect possible trends, and evaluate whether the variables supported the assumptions needed for regression modeling.

The first visualization was a scatterplot of **goals scored (scored)** versus **Expected Goals (xG)**. The scatterplot showed a clear positive relationship between the two variables, indicating that teams with higher expected goals generally scored more goals. As xG increased, the number of goals scored also tended to increase, suggesting that shot quality is strongly associated with offensive output. The plot also showed horizontal bands across the graph because goals scored is a count variable that can only take whole number values such as 0, 1, 2, or 3 goals. A fitted regression line further supported the positive linear trend between xG and goals scored.

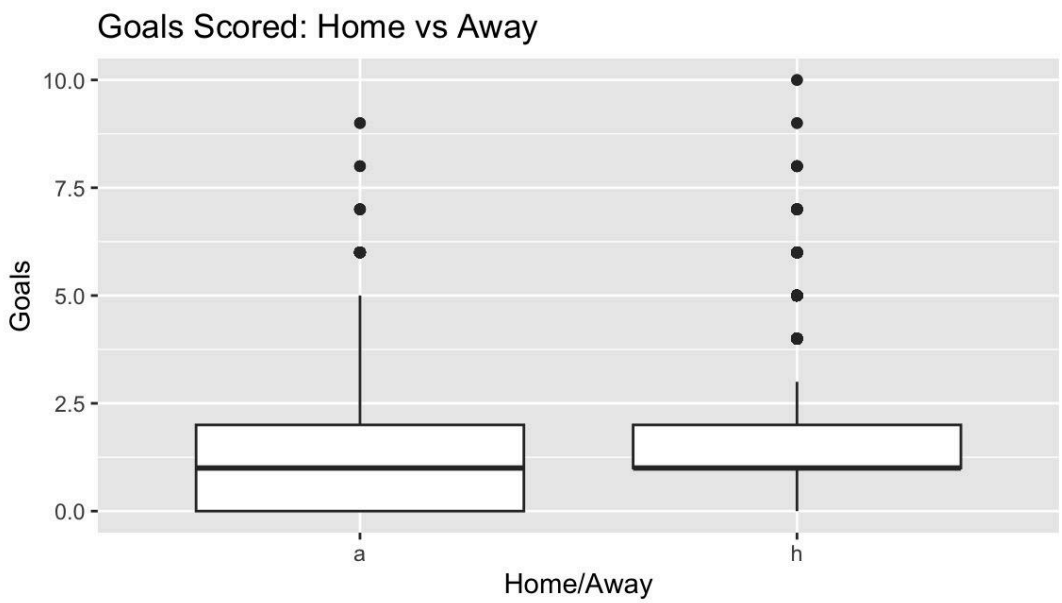


A boxplot was also created to compare **xG values across match results**, including wins, losses, and draws. The results showed that teams that won matches generally had higher xG values

compared to teams that drew or lost. The median xG for winning teams was noticeably higher, suggesting that creating higher quality scoring opportunities is strongly associated with match success. In contrast, teams that lost matches tended to have lower xG values, indicating fewer dangerous attacking opportunities.



Another boxplot compared **goals scored between home and away teams**. The visualization showed that home teams scored slightly more goals on average compared to away teams. The median number of goals for home teams was higher, and home teams also showed slightly greater offensive consistency. This finding supports the idea of home field advantage in professional soccer, where familiar surroundings and crowd support may positively influence attacking performance.



Overall, the exploratory analysis revealed meaningful relationships between xG, goals scored, match outcomes, and home advantage. These findings provided strong initial evidence that expected goals may be an important predictor of actual goals scored and justified the use of regression based modeling in this study.

Statistical Models

I began doing a simple linear regression model that was used to examine the relationship between **expected goals (xG)** and **goals scored (scored)**. The purpose of this model was to determine whether xG alone could significantly predict offensive performance in professional soccer matches.

The results showed a significant positive relationship between expected goals and goals scored. As xG increased, the number of goals scored also tended to increase. The model produced a statistically significant slope coefficient (**$p < 0.001$**), indicating that xG was a meaningful predictor of goal scoring. These findings provide initial evidence that expected goals is strongly

associated with actual scoring performance and serve as a useful baseline for more advanced statistical models.

After the simple linear regression, a multiple linear regression model was developed to examine whether additional team performance variables could improve goal prediction. The model included **xG_diff**, **xGA_diff**, and **npxGD** as predictor variables to capture both offensive and defensive performance. The results showed that the multiple regression model explained substantially more variation in goals scored compared to the simple regression model. The model produced an **R^2 of approximately 0.79**, indicating strong predictive performance. In addition, all major predictor variables showed meaningful contributions, suggesting that combining multiple performance metrics improves the accuracy of goal prediction in professional soccer.

A polynomial regression model was used to examine whether the relationship between expected goals and goals scored followed a nonlinear pattern. Unlike simple linear regression, polynomial regression allows the relationship between variables to change across different levels of xG rather than follow a single straight line. This approach can better capture scoring patterns when offensive performance does not increase at a constant rate. The results showed that the polynomial model explained approximately **58.4%** of the variation in goals scored (**$R^2 \approx 0.584$**), indicating a moderate fit. These findings suggest that nonlinear relationships may exist between expected goals and actual goal scoring in soccer.

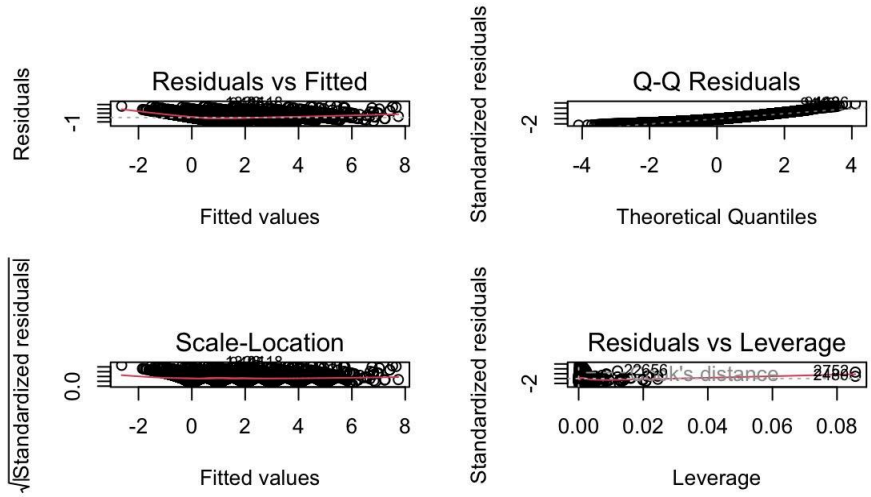
A regression model was developed by adding pressing related variables to examine whether tactical and defensive behaviors could improve goal prediction. The model included variables

such as **ppda_coef**, **oppda_coef**, **ppda_att**, **ppda_def**, **oppda_att**, and **oppda_def**, which measure defensive intensity and opponent pressure during matches. By including these tactical variables, the model captured additional aspects of team performance beyond offensive statistics alone. The results showed that the model explained approximately **59.0%** of the variation in goals scored (**$R^2 \approx 0.590$**). These findings suggest that pressing behavior provides useful information and contributes to a better understanding of goal scoring performance in soccer.

A full multiple regression model was developed to examine how offensive, defensive, and tactical variables jointly predict goals scored in professional soccer. The model included **xG_diff**, **xGA_diff**, **npxGD**, **ppda_coef**, **oppda_coef**, **ppda_att**, **ppda_def**, **oppda_att**, and **oppda_def** as predictor variables. The results showed that the model explained approximately **79.7%** of the variation in goals scored (**$R^2 \approx 0.797$**), which was the highest among all regression models tested. This indicates that the full model provided the best overall fit. However, **ppda_def** and **oppda_def** were not statistically significant, suggesting they may have limited predictive value.

Diagnostic plots were examined to assess model assumptions. The residual plots showed no major violations of linearity or constant variance, while the Q–Q plot suggested approximate normality of residuals. In addition, leverage diagnostics indicated no highly influential observations that would substantially affect model estimates.

Diagnostic Plots for the Full Multiple Regression Model



Poisson regression is specifically designed for count outcomes and estimates the expected number of goals scored in each match. The model was used to examine whether the predictor variables remained significant when accounting for the count nature of the response variable. The results showed that the main predictor variables remained statistically significant and had positive relationships with goals scored, confirming that expected goals and team performance metrics are strong predictors of scoring outcomes in professional soccer.

A negative binomial regression model was used to test for possible overdispersion in the goal scoring data, which occurs when the variance is larger than the mean. Since soccer goals are count data and may show greater variability than assumed under a Poisson distribution, this model provided a useful alternative. The results were similar to those from the Poisson regression model, with the main predictor variables remaining significant. These findings suggest that both count models provide consistent and reliable estimates for predicting goals scored.

Ridge and LASSO regression were used as regularization techniques to improve model stability and reduce the risk of overfitting when multiple predictor variables were included. These methods apply penalties to regression coefficients and help identify the most important variables in the model. The results showed that the coefficient estimates remained stable under ridge regression, indicating consistent predictor effects across the model. LASSO regression further simplified the model by shrinking weaker predictors toward zero, effectively removing variables with limited predictive contribution.

Results and Interpretation

The results of this study showed that **expected goals (xG)** is a strong and statistically significant predictor of goals scored in professional soccer matches. The simple linear regression model confirmed a positive relationship between xG and goals scored, providing initial evidence that teams creating higher quality scoring opportunities generally score more goals.

The multiple regression models outperformed the simple regression model by including additional offensive and defensive variables. The full multiple regression model produced the best overall fit, with an **R^2 of approximately 0.797**, explaining nearly 80% of the variation in goals scored.

The Poisson and negative binomial regression models produced results similar to the linear models, confirming that the main predictor variables remained significant when accounting for the count nature of goals. Pressing related variables slightly improved model performance, showing that tactical behavior contributes useful information. However, variables such as **ppda_def** and **oppda_def** showed limited predictive value in the final model.

Table: Regression Coefficients and Model Fit Statistics Across Regression Models

	Simple Linear	Multiple Linear	Polynomial	Pressing Model	Full Model	Log Model	Poisson	Negative Binomial
npxGD	0.514***	0.516***			0.511***	0.208***	0.323***	0.323***
	(0.006)	(0.003)			(0.003)	(0.001)	(0.004)	(0.004)
xG_diff		-0.967***	-0.871***	-0.977***	-0.963***	-0.397***	-0.520***	-0.520***
		(0.004)	(0.006)	(0.005)	(0.004)	(0.002)	(0.005)	(0.005)
xGA_diff		0.019***			0.024***	0.006***	0.010	0.010
		(0.004)			(0.004)	(0.002)	(0.006)	(0.006)
I(xG_diff^2)			0.164***					
			(0.003)					
ppda_coef				-0.020***	0.006***			
				(0.001)	(0.001)			
oppda_coef				0.035***	0.005***			
				(0.001)	(0.001)			
Num.Obs.	24580	24580	24580	24580	24580	24580	24580	24580
R2	0.248	0.794	0.584	0.590	0.797	0.730		
R2 Adj.	0.248	0.794	0.584	0.590	0.797	0.730		

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

The table above summarizes the coefficient estimates and model fit statistics across all regression models tested in this study. The full multiple regression model produced the strongest predictive performance, achieving an R^2 of approximately 0.797, followed closely by the multiple linear regression model ($R^2 \approx 0.794$). In contrast, the simple linear regression model produced the lowest explanatory power ($R^2 \approx 0.248$). Across nearly all models, the expected goals related variables, particularly **xG_diff** and **npxGD**, remained statistically significant predictors of goals scored, demonstrating the strong predictive value of expected goals metrics in professional soccer.

Discussion

The findings of this study show that **expected goals (xG)** is a valuable tool for understanding offensive performance in professional soccer. Since xG measures the quality of

scoring opportunities rather than only the final score, it provides deeper insight into how well a team creates chances during a match. The strong relationship between xG and goals scored suggests that teams creating higher quality chances generally score more goals.

However, xG does not perfectly predict goal scoring because soccer contains randomness and unpredictability. Factors such as finishing ability, goalkeeper performance, defensive pressure, and game situations can influence whether a scoring opportunity becomes a goal. Some players consistently perform above their expected goals because of exceptional finishing skill. For example, **Mohamed Salah** has regularly outperformed his xG in professional competition, showing how elite players can finish chances more efficiently than expected. At the same time, goalkeepers can prevent high quality chances from becoming goals through exceptional saves. Match conditions, luck, and important game moments also affect scoring outcomes. These findings show that xG is a powerful analytical tool, but soccer will always contain elements that statistics cannot fully explain.

Limitations

Although this study produced strong results, several limitations should be considered when interpreting the findings. First, the dataset was obtained from **Understat**, which mainly covers major European professional leagues. As a result, the findings may not fully represent lower divisions, youth competitions, or leagues outside Europe where playing styles and match conditions may differ.

Second, this study used **match level data** rather than player tracking data. While match level statistics provide useful team performance measures, they do not capture real time player

movement, defensive positioning, or off the ball actions that may influence goal scoring. Including tracking data could provide a deeper understanding of tactical behavior.

In addition, important external factors such as player injuries, weather conditions, fatigue, and fixture congestion were not included in the analysis. These variables may affect team performance and scoring outcomes during a season. Finally, detailed tactical context such as formation changes, substitutions, and in game strategy adjustments was not available in the dataset. Including these factors in future research may improve model accuracy and provide a more complete understanding of goal scoring performance in soccer.

Future Research

Future research can expand this study by including more advanced data and modeling techniques in soccer analytics. One important improvement would be the use of player tracking data, which captures real time player movement, positioning, and defensive pressure during matches. This could provide deeper tactical insights that are not available in match level data alone. Machine learning methods such as random forests, gradient boosting, and neural networks could also be used to improve prediction accuracy and identify more complex patterns. In addition, future studies could examine **player level expected goals (xG)** to evaluate individual finishing ability. Team specific models could also be developed to better understand how tactical styles influence goal scoring performance across different clubs.

Conclusion

This project demonstrates that **expected goals (xG)** is a statistically significant predictor of actual goals scored in professional soccer. Across multiple statistical approaches, including

simple linear regression, multiple regression, polynomial regression, Poisson regression, negative binomial regression, ridge regression, and LASSO regression, xG consistently showed strong predictive power. The results revealed a clear positive relationship between expected goals and goals scored, supporting the idea that teams creating higher quality scoring opportunities are more likely to score goals.

The inclusion of additional offensive, defensive, and pressing variables further improved model performance, with the full multiple regression model providing the best overall fit. These findings show that combining xG with other performance metrics can provide a more complete understanding of goal scoring in soccer.

Although soccer remains influenced by randomness, finishing ability, goalkeeper performance, and match context, xG continues to provide valuable insight beyond traditional statistics. Overall, this study shows that expected goals is an effective analytical tool for evaluating team and player performance while helping coaches, analysts, and organizations make more informed decisions in modern soccer.

References:

Mead, J., O'Hare, A., & McMenemy, P. (2023). Expected goals in football: Improving model performance and demonstrating value. *PLOS ONE, 18*(4), e0282295. https://doi.org/10.1371/journal.pone.0282295

Hewitt, J. H., & Karakuş, O. (2023). A machine learning approach for player and position adjusted expected goals in football (soccer). *Franklin Open, 4*, 100034. https://www.sciencedirect.com/science/article/pii/S2773186323000282?via%3Dihub

Lang, S., Wild, R., Isenko, A., & Link, D. (2022). Predicting the in-game status in soccer with machine learning using spatiotemporal player tracking data. *Scientific Reports, 12*, 16291. https://www.nature.com/articles/s41598-022-19948-1

The Coaches' Voice. (2022, June 22). *Expected goals: Explained.* Coaches' Voice. https://learning.coachesvoice.com/cv/expected-goals-xg-explained/